



BRAC University

Department of Mathematics and Natural Sciences

LECTURE ON

Real Analysis (MAT221)

Riemann Integral

Upper and Lower Sums, Riemann Integrability

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CONDUCTED BY

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Historical and Conceptual Motivation

“The integral was not born as a limit—it was born as area.”

In elementary calculus, integration is often introduced as the inverse operation of differentiation. That approach is historically inaccurate and mathematically insufficient.

The founders of calculus—Newton and Leibniz—viewed integration primarily through antiderivatives. However, this immediately leads to a serious limitation:

❓ Problem

Show that the function

$$h(x) = \begin{cases} 1, & 0 \leq x < 1, \\ 2, & 1 \leq x \leq 2, \end{cases}$$

has no antiderivative on $[0, 2]$.

Despite this, it is intuitively clear that the area under h from 0 to 2 should be 3. Thus, **area must be defined independently of derivatives**.

This philosophical shift—separating integration from differentiation—was made rigorous by Cauchy and later refined by Riemann around 1850
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From Geometry to Approximation

The geometric idea behind integration is approximation:

Approximate the region under a curve by rectangles, and refine the approximation.

Each approximation is crude, but refinement improves accuracy. The mathematical challenge is to turn this idea into a precise definition that depends only on properties of real numbers—especially completeness.

Bounded Functions on Closed Intervals

Bounded Function

A function $f : [a, b] \rightarrow \mathbb{R}$ is called **bounded** if there exists $M > 0$ such that

$$|f(x)| \leq M \quad \text{for all } x \in [a, b].$$

Boundedness is not optional. If f is unbounded on any subinterval, then no finite rectangular approximation is possible.

Theorem

Every continuous function on a closed interval $[a, b]$ is bounded.

This is a direct consequence of compactness and the Extreme Value Theorem.

Partitions of an Interval

Partition

A **partition** P of $[a, b]$ is a finite ordered set

$$P = \{x_0, x_1, \dots, x_n\}$$

such that

$$a = x_0 < x_1 < \dots < x_n = b.$$

Each partition induces subintervals $[x_{i-1}, x_i]$.

Norm (Mesh)

The **norm** of a partition P is

$$\|P\| = \max_{1 \leq i \leq n} (x_i - x_{i-1}).$$

A small norm corresponds to a fine partition. In the limit, we want $\|P\| \rightarrow 0$.

Supremum and Infimum on Subintervals

Definition

Let f be bounded function on $[a, b]$ and let $[x_{i-1}, x_i]$ be a subinterval of the interval. Define

$$M_i = \sup\{f(x) : x \in [x_{i-1}, x_i]\}.$$

$$m_i = \inf\{f(x) : x \in [x_{i-1}, x_i]\}.$$

These quantities always exist because:

- f is bounded,
- $\mathbb{R}$ is complete.

They represent the best possible overestimate and underestimate of f on each subinterval.

Upper and Lower Sums

Lower Sum

Let f be bounded function on $[a, b]$ and let $P = \{x_0, x_1, \dots, x_n\}$ be a partition of $[a, b]$. Define

$$L(f, P) = \sum_{i=1}^n m_i(x_i - x_{i-1}).$$

Upper Sum

Let f be bounded function on $[a, b]$ and let $P = \{x_0, x_1, \dots, x_n\}$ be a partition of $[a, b]$. Define

$$U(f, P) = \sum_{i=1}^n M_i(x_i - x_{i-1}).$$

Theorem

For any partition P ,

$$L(f, P) \leq U(f, P).$$



Geometrically:

- Lower sums inscribe rectangles below the graph,
- Upper sums circumscribe rectangles above the graph.

Refinement and Monotonicity of Sums

Refinement

A partition Q is a refinement of P if $P \subseteq Q$.



Theorem

If Q refines P , then



$$L(f, P) \leq L(f, Q) \leq U(f, Q) \leq U(f, P).$$

Thus:

- lower sums increase under refinement,
- upper sums decrease under refinement.

This squeezing behavior is the key to integrability

Upper and Lower Integrals

Instead of taking limits of sums, Riemann used infima and suprema.

Upper Integral

$$\overline{\int_a^b} f = \inf \{U(f, P) : P \text{ partition of } [a, b]\}.$$





Lower Integral

$$\int_a^b f = \sup\{L(f, P) : P \text{ partition of } [a, b]\}.$$



Theorem

For every bounded function,

$$\int_a^b f \leq \overline{\int_a^b f}.$$



Riemann Integrability

Riemann Integrable

A bounded function $f : [a, b] \rightarrow \mathbb{R}$ is **Riemann integrable** if 

$$\int_a^b f = \overline{\int_a^b f}.$$

The common value is denoted

$$\int_a^b f(x) \, dx.$$

Cauchy Criterion

f is Riemann integrable if and only if for every $\varepsilon > 0$ there exists a partition P such that

$$U(f, P) - L(f, P) < \varepsilon.$$

This theorem is the practical test for integrability.

Continuity Implies Integrability

Theorem

If f is continuous on $[a, b]$, then f is Riemann integrable.



Functions with Discontinuities

Not all integrable functions are continuous.

❓ Problem

Define

$$f(x) = \begin{cases} 1, & x \neq 1, \\ 0, & x = 1. \end{cases}$$

Then f is Riemann integrable on $[0, 2]$ and

$$\int_0^2 f(x) dx = 2.$$

A finite number of discontinuities does not affect area.

A Pathological Counterexample

Dirichlet Function

$$f(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \notin \mathbb{Q}. \end{cases}$$


Theorem

The Dirichlet function is not Riemann integrable on $[0, 1]$.



Lebesgue's Criterion (Advanced Real Analysis)

Lebesgue Criterion

A bounded function is Riemann integrable if and only if its set of discontinuities has measure zero. 

This theorem precisely describes the power and limitation of Riemann integration.

Thank You!

We'd love your questions and feedback.

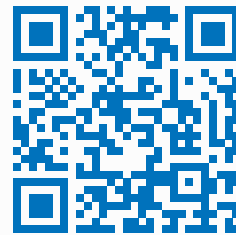
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References

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- [2] Terence Tao, *Analysis I*, 3rd Edition, Texts and Readings in Mathematics, Hindustan Book Agency, 2016.